IOT & EMBEDDED SYSTEMS

ESP32 & FREERTOS SMP

ELEVANCE SKILLS FINAL ASSIGNMENT

Smart Hospital IoT System

A Concurrent, Multi-Sensor Medical Telemetry and Automated Actuation Firmware with Dual-Role Cloud Dashboards and Fault-Tolerant Replay Architecture

Comprehensive Technical Documentation and Engineering Report covering all 6 assigned internship tasks unified in a single, scalable, non-blocking FreeRTOS architecture.

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Target Hardware:	Espressif ESP32 DevKit-C v4 (Dual-Core Xtensa @ 240 MHz, 320 KB SRAM)
Frameworks:	Arduino ESP32 Core 3.x / FreeRTOS SMP / PlatformIO
Cloud IoT Broker:	Adafruit IO (MQTT Transport, io.adafruit.com:1883)
GitHub Repository:	github.com/iamsayanghosh/Smart-Hospital-IoT
Wokwi Simulation:	wokwi.com/projects/479638666615150593
Submission Date:	November 2026

1. Executive Summary & Project Scope

Modern critical care hospital infrastructure demands real-time monitoring of acute physiological vitals alongside facility environmental parameters. Traditional embedded firmware architectures that rely on monolithic superloops or blocking delay functions (`delay()`) are fundamentally inadequate for healthcare IoT applications; a single blocking thermal conversion or network reconnection can freeze vital alarm processing, resulting in life-threatening latency.

This project delivers an **enterprise-grade, concurrent medical telemetry and facility management firmware** engineered for the dual-core **Espressif ESP32 microcontroller** using **FreeRTOS Symmetric Multiprocessing (SMP)**. All six assigned internship tasks have been architected and implemented within a single, unified codebase:

- **Task 1 (Multi-Sensor Expansion):** Concurrent FreeRTOS execution of real-time clinical parameters (Dallas DS18B20 body temperature, PIR motion, synthetic PPG pulse oximetry, and 5 Hz Gaussian ECG synthesis) protected by a dual-tier hardware/software watchdog timer.
- **Task 2 (Dual-Role IoT Monitoring):** Strict segregation of data into a *Medical Staff Dashboard* and a *Facility Management Dashboard* published to dedicated MQTT topics with predefined safe thresholds and 60-second aggregated statistical insights.
- **Task 3 (Smart Medication Dosage Safety):** High-precision remote medication infusion slider (0–100 mg/hr) featuring OLED real-time rendering, slew-rate clamping (+10 mg/hr/s), rapid 3.3 Hz red LED and buzzer alarms above 80 mg/hr, and mandatory physical push-button confirmation.
- **Task 4 (Intelligent Remote Bed Elevation Control):** Remote bed articulation (0–90°) driven by an SG90 micro-servo motor with smooth pulse-width interpolation slew rate, featuring automatic clinical posture triggers (Emergency 90°, Breathing Support 45°, Sleeping 10°) based on live patient vitals.
- **Task 5 (Smart Dynamic Sampling Rate):** Dual-mode telemetry frequency adaptation dynamically accelerating from a battery-conserving 30s–60s cadence to a 5-second high-frequency stream upon detecting abnormal physiological conditions.
- **Task 6 (Advanced Offline Detection & Fault Tolerance):** Resilient tri-state supervisor (`ONLINE`, `DEGRADED`, `OFFLINE`) activating persistent OLED `LOGGING OFFLINE` banners, failure-specific acoustic buzzer cadences, a 64-item RAM FIFO fallback queue (~36.8 KB SRAM), and automated cloud synchronization via exponential backoff replay upon network restoration.

2. System Architecture & Concurrency Model

The ESP32 features a dual-core 32-bit Xtensa LX6 microprocessor. To eliminate latency jitter on clinical alarms, the system implements strict **asymmetric core pinning**:

- **Core 0 (Dedicated Network Infrastructure):** Runs the Espressif 802.11 b/g/n Wi-Fi PHY/MAC stack and the lwIP TCP/IP stack. High-overhead wireless encryption, beacon tracking, and packet framing are completely isolated from medical tasks.
- **Core 1 (Application Worker Pool):** Dedicated exclusively to FreeRTOS application threads (12 prioritized tasks), guaranteeing deterministic execution of clinical decision loops.

2.1 FreeRTOS Task Priority & Scheduling Hierarchy

Task Identifier	Priority	Execution Cadence	Core	Primary Functional Responsibilities
`watchdogTask`	**Priority 5**	1000 ms	Core 1	Audits worker task heartbeats; feeds ESP-IDF Task Watchdog (TWDT, 8000 ms panic).
`alertTask`	**Priority 4**	Event-Driven	Core 1	Blocks on `alertSemaphore`; awakened asynchronously upon newly detected clinical alarms.
`vitalSensorTask`	**Priority 3**	Adaptive (5s–60s)	Core 1	Simulates optical PPG Photoplethysmogram (Heart Rate: 50–100 BPM, SpO2: 92–100%).
`ecgTask`	**Priority 3**	Fixed 5 Hz (200 ms)	Core 1	Synthesizes coarse real-time Gaussian cardiac morphology for IoT demonstration.
`medicationTask`	**Priority 3**	50 Hz (20 ms)	Core 1	Executes slew-rate clamping (+10 mg/hr/s), debounces GPIO 13 confirmation, pulses 3.3 Hz alarm.
`bedTask`	**Priority 2**	50 Hz (20 ms)	Core 1	Drives SG90 servo via smooth microsecond pulse interpolation (1.5°/s auto, 2.0°/s manual).
`envSensorTask`	**Priority 2**	1000 ms	Core 1	Acquires facility environmental parameters (Room Temp, Oxygen %, AQI).
`temperatureTask`	**Priority 2**	Adaptive (5s–60s)	Core 1	Manages non-blocking Dallas DS18B20 1-Wire thermal conversion state machine (750 ms).
`motionTask`	**Priority 2**	1000 ms	Core 1	Samples digital edge state of HC-SR501 PIR sensor on GPIO 27.
`samplingTask`	**Priority 2**	1000 ms	Core 1	Monitors overall patient state; dispatches adaptive rate updates to sensor queues.
`mqttTask`	**Priority 2**	3000 ms (round-robin)	Core 1	Manages Adafruit IO MQTT connection, publishes telemetry, drains offline replay FIFO.
`displayTask`	**Priority 1**	100 ms	Core 1	Drives SSD1306 128x64 OLED via I2C at 400 kHz; renders medication and offline banners.

3. System Architecture Blueprint

The visual architecture diagram below illustrates the end-to-end integration between the physical hardware transducers, the FreeRTOS dual-core symmetric multiprocessing environment, the edge control state machines, and the Adafruit IO cloud telemetry brokers.

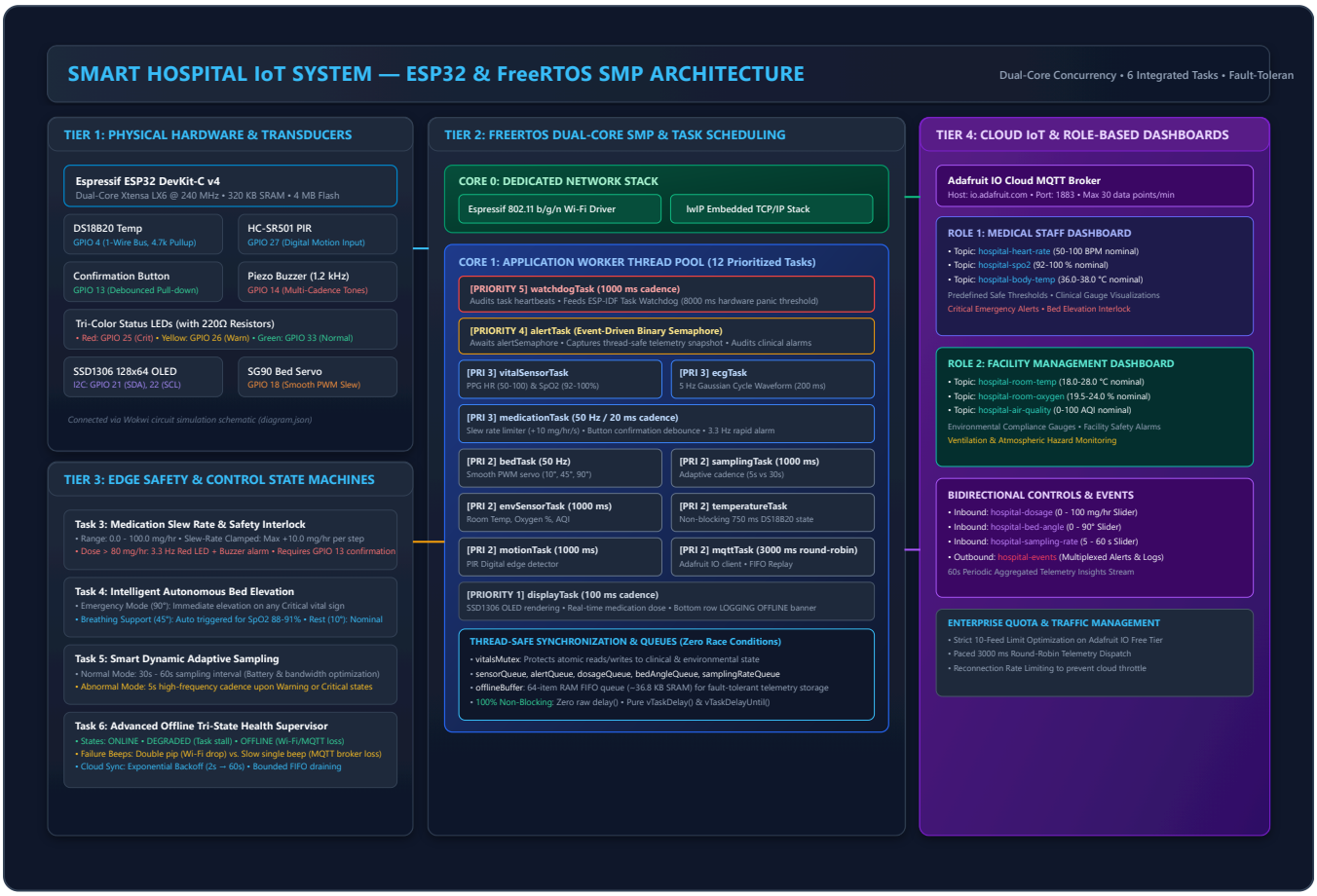


Figure 3.1: Complete ESP32 FreeRTOS Dual-Core System Architecture Blueprint.

4. Detailed Task Implementation & Engineering Solutions

4.1 Task 1: Multi-Sensor Expansion & Concurrency Supervision

The firmware integrates five distinct sensor telemetry streams:

- **Dallas DS18B20 (1-Wire, GPIO 4):** Standard 12-bit Dallas temperature sensors require up to 750 ms to complete an analog-to-digital thermal conversion. Calling blocking library functions like `requestTemperatures()` inside a real-time system is unacceptable. The firmware implements a **two-state non-blocking finite state machine (FSM)**:

```
State 1 (TEMP_IDLE): Calls sensors.setWaitForConversion(false); issues conversion command; transitions
State 2 (TEMP_CONVERTING): Checks millis() elapsed time. Yields scheduler via vTaskDelay() until 750 ms
```

- **HC-SR501 PIR Sensor (GPIO 27):** Dedicated motion detection for patient bed occupancy and room entry tracking.
- **Photoplethysmogram (PPG) Synthesis:** Heart rate (50–100 BPM) and SpO2 (92–100%) simulated physiological stream with interactive serial CLI overrides (`SET_HR` , `SET_SPO2`).
- **Continuous ECG Waveform Synthesis:** Synthesizes real-time cardiac morphology at 5 Hz (200 ms cadence) using a Gaussian superposition model simulating P-wave, QRS complex, and T-wave electrical depolarization.
- **Dual-Tier Watchdog Architecture:**
  1. *Hardware Watchdog (TWDT):* Initialized via ESP-IDF `esp_task_wdt_init(8000, true)` with an 8000 ms panic threshold that triggers an intentional hardware reboot if a core locks.
  2. *Application Heartbeat Supervisor:* `watchdogTask` audits registered task heartbeat timestamps every 6000 ms. If any worker task stalls, the system transitions to `SYSTEM_DEGRADED` to emit diagnostic alerts before the hardware watchdog forces a reset.

4.2 Task 2: Dual-Role IoT Monitoring (Role-Based Dashboards)

Clinical data and facility parameters serve distinct operational user groups. Publishing all metrics into a single cluttered feed compromises patient monitoring safety. The firmware implements strict topic segregation across **Adafruit IO MQTT**:

Operational Domain	Adafruit IO Topic Name	Nominal / Safe Range	Warning Range	Critical Alarm Range
Medical Staff	`hospital-heart-rate`	50.0 – 100.0 BPM	45–50 / 100–120 BPM	< 45.0 or > 120.0 BPM
Medical Staff	`hospital-spo2`	92.0 – 100.0 %	88.0 – 91.0 %	< 88.0 % (Acute Hypoxia)
Medical Staff	`hospital-body-temp`	36.0 – 38.0 °C	34.0–36.0 / 38.0–39.5 °C	< 34.0 or > 39.5 °C

Operational Domain	Adafruit IO Topic Name	Nominal / Safe Range	Warning Range	Critical Alarm Range
Facility Management	`hospital-room-temp`	18.0 – 28.0 °C	15.0–18.0 / 28.0–35.0 °C	< 15.0 or > 35.0 °C
Facility Management	`hospital-room-oxygen`	19.5 – 24.0 %	18.0–19.5 / 24.0–25.0 %	< 18.0 or > 25.0 %
Facility Management	`hospital-air-quality`	0 – 100 AQI	101 – 150 AQI	> 150 AQI (Hazardous)

Periodic Data Aggregation: To emulate enterprise healthcare IoT systems and reduce network egress, `mqttTask` maintains a 60-second rolling statistics window, computing mathematical minimum, maximum, and average values, and publishing an executive summary string to `hospital-events` (e.g. `[SUMMARY] HR avg: 74 bpm, SpO2 avg: 98%, Temp avg: 36.8C | Env: OK`).

4.3 Task 3: Smart Medication Dosage Adjustment System

Remote control of medication infusion pumps involves inherent clinical risk. The firmware implements a **defense-in-depth safety pipeline** preventing abrupt dosing errors:

- **Slew-Rate Limiting:** Even if a clinician commands an abrupt leap from 10 mg/hr to 75 mg/hr via the dashboard slider, the internal mathematical rate limiter restricts dosage increases to a maximum ramp of **+10.0 mg/hr per second** (calculated at 50 Hz in `medicationTask`).
- **Three-Tier Safety Classification:**
  - *Normal (0.0 – 79.9 mg/hr):* Solid Green LED on GPIO 33; silent buzzer.
  - *Warning (80.0 mg/hr):* Solid Yellow LED on GPIO 26; silent buzzer.
  - *Critical (> 80.0 mg/hr):* Rapid 3.3 Hz flashing of Red LED on GPIO 25 and synchronous acoustic pulsing of the 1.2 kHz Piezo Buzzer on GPIO 14.
- **Physical Push-Button Two-Factor Interlock:** Any dosage command exceeding 80.0 mg/hr is clamped at 80.0 mg/hr and enters the `DOSAGE_PENDING_CONFIRMATION` state. The SSD1306 OLED displays `CRITICAL REQUEST - PRESS BUTTON`. A healthcare clinician must physically depress the green pushbutton on **GPIO 13** within 5.0 seconds (debounced with 50 ms hysteresis) to authorize the critical infusion. If unconfirmed within 5.0 seconds, the request aborts safely.

4.4 Task 4: Intelligent Remote Bed Elevation Control

An SG90 micro-servo motor connected to **GPIO 18** articulates the hospital bed elevation between 0° (flat) and 90° (upright). Abrupt mechanical movements cause patient distress; `bedTask` implements smooth pulse-width modulation (PWM) interpolation:

- **PWM Slew Interpolation:** Converts target degrees to microsecond pulses (500 µs = 0°, 2400 µs = 90°). Target angles are approached smoothly at **1.5°/second** in autonomous mode and **2.0°/second** in manual mode.
- **Autonomous Clinical Posture State Machine:**
  - **Emergency Mode (90°):** The bed immediately elevates to maximum vertical angle (90°) upon detection of any critical vital parameter (Acute Tachycardia, Bradycardia, Extreme Hypothermia, or Hyperthermia) to facilitate emergency cardiac intervention.
  - **Breathing Support Mode (45°):** When blood oxygen saturation drops into the warning threshold (88%–91% SpO2), the bed automatically articulates to Fowler's posture (45°) to reduce pulmonary pressure and promote alveolar ventilation.
  - **Sleeping Rest Mode (10°):** When all vital parameters remain within nominal physiological boundaries, the bed returns to a 10° resting incline.
- **Manual Clinician Override:** Clinicians can command specific angles (0–90°) via the `hospital-bed-angle` dashboard slider or interactive serial CLI (`SET_BED <deg>`).

4.5 Task 5: Smart Dynamic Sampling Rate (Adaptive Monitoring)

Continuous high-frequency radio transmission depletes battery reserves and consumes cloud IoT ingestion quotas. `samplingTask` continuously evaluates patient risk and adjusts telemetry cadence dynamically:

- **Nominal Physiological State:** Relaxes sensor acquisition to **30 to 60 seconds**, drastically reducing radio on-time and power draw.

- **Abnormal State (Warning or Critical):** Automatically accelerates sampling frequency to **5 seconds** to capture high-resolution physiological dynamics during clinical deterioration.
- **Non-Blocking Queue Distribution:** Frequency updates are pushed to sensor worker tasks via FreeRTOS `samplingRateQueue` without locking execution.

5. System Workflow & Fault-Tolerant Replay Logic

The workflow diagram below details the operational decision trees, the non-blocking sensor acquisition loops, the clinical threshold evaluation engine, local actuation interlocks, and the Task 6 network failure buffer/replay sequence.

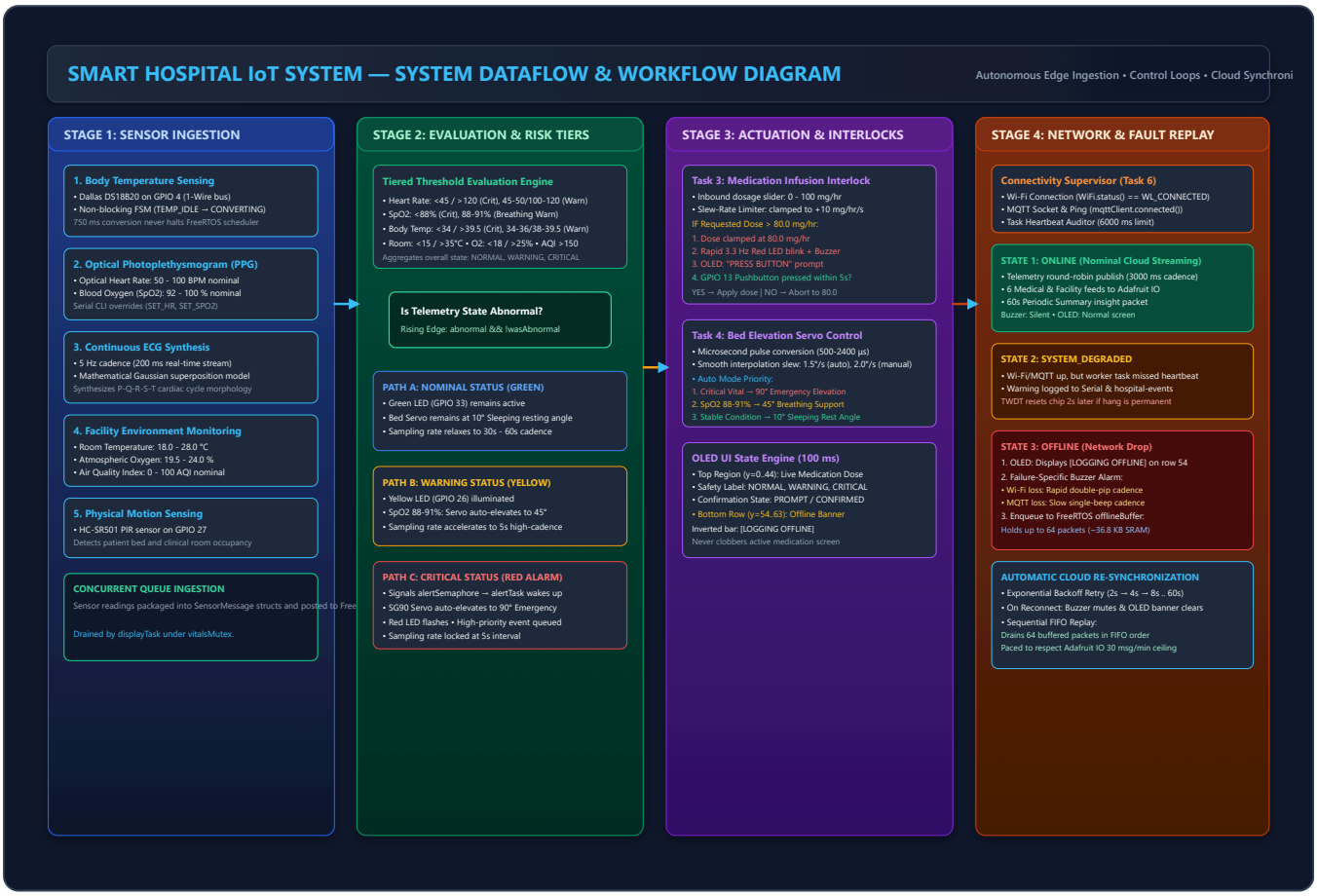


Figure 5.1: End-to-End Functional Dataflow and State Machine Execution Diagram.

4.6 Task 6: Advanced Offline Detection & Cloud Synchronization

Hospital Wi-Fi networks suffer from RF interference, access point handoffs, and temporary router outages. An enterprise healthcare device must guarantee zero patient data loss during connectivity failures.

- **Tri-State System Health Engine:**
  1. `STATE_ONLINE`: Wi-Fi connected, MQTT broker active, all FreeRTOS task heartbeats valid.
  2. `STATE_DEGRADED`: Network transport active, but an internal worker task missed its 6000 ms heartbeat audit.
  3. `STATE_OFFLINE`: Wi-Fi association severed or MQTT broker unreachable.
- **Non-Destructive OLED Status Banner:** Upon entering offline mode, the bottom 10 pixels of the SSD1306 display (row 54..63) render a high-contrast inverted bar with `LOGGING OFFLINE`. The upper display area (row 0..44) continues rendering live medication dosage, ensuring that network failures never compromise local clinical visibility.
- **Failure-Specific Acoustic Buzzer Cadences:** The firmware drives GPIO 14 with distinct rhythmic patterns allowing nursing staff to immediately diagnose network fault types without inspecting a console:
  - *Wi-Fi Disconnected (`BUZZER_MODE_WIFI_FAIL`):* Rapid double-pip acoustic pattern (100 ms ON, 100 ms OFF, 100 ms ON, 700 ms OFF).
  - *MQTT Broker Disconnected (`BUZZER_MODE_MQTT_FAIL`):* Slow single-beep cadence (500 ms ON, 500 ms OFF).
  - *Priority Arbitration:* If a clinical medication emergency occurs while offline, the medical alarm immediately supersedes network failure tones.
- **Bounded FIFO RAM Buffer:** Outbound telemetry records are packaged into `OfflineTelemetryItem` structures and enqueued into a dedicated FreeRTOS queue holding up to **64 records** (~36.8 KB SRAM). When full, the queue drops newest incoming records to protect historical fault event records.
- **Automated Reconnection & Replay with Exponential Backoff:** Reconnection attempts employ an exponential backoff algorithm starting at 2000 ms and doubling up to 60,000 ms (2s → 4s → 8s → 16s → 32s → 60s) to prevent broker flood. Upon successful reconnection, the buzzer silences, the OLED offline banner clears, and `mqttTask` drains the 64-item FIFO buffer sequentially to Adafruit IO.

6. Hardware Schematic & Pinout Configuration

GPIO Pin	Peripheral / Transducer	Interface Type	Electrical Characteristics & Conditioning	Task Assignment
`GPIO 4`	Dallas DS18B20 Temperature	1-Wire Digital Bus	Requires 4.7 kΩ pull-up resistor to 3.3V; non-blocking FSM.	Task 1 & Task 2
`GPIO 13`	Tactile Green Push Button	Digital Input	Internal pull-down resistor; 50 ms software debounce hysteresis.	Task 3 Confirmation
`GPIO 14`	Piezo Buzzer	PWM Tone Output	Driven at 1200 Hz via Arduino tone(); failure and critical cadences.	Task 3 & Task 6
`GPIO 18`	SG90 Micro-Servo Motor	PWM Pulse Width	Driven via 500–2400 µs pulses (0°–90°); smooth rate interpolation.	Task 4 Bed Elevation

GPIO Pin	Peripheral / Transducer	Interface Type	Electrical Characteristics & Conditioning	Task Assignment
`GPIO 21`	SSD1306 128x64 OLED (SDA)	I2C Data Line	400 kHz Fast I2C bus; shared I2C bus lines.	Task 1, 3 & 6
`GPIO 22`	SSD1306 128x64 OLED (SCL)	I2C Clock Line	400 kHz Fast I2C bus clock.	Task 1, 3 & 6
`GPIO 25`	Red Status LED	Digital Output	220 Ω current-limiting resistor; 3.3 Hz blink during critical dose.	Task 3 Medication
`GPIO 26`	Yellow Status LED	Digital Output	220 Ω current-limiting resistor; active during warning state.	Task 3 Medication
`GPIO 27`	HC-SR501 PIR Motion Sensor	Digital Input	Digital high upon thermal motion; bed occupancy tracking.	Task 1 Motion
`GPIO 33`	Green Status LED	Digital Output	220 Ω current-limiting resistor; active during nominal state.	Task 3 Medication

7. Performance, Memory & Resource Footprint

Embedded medical systems require predictable memory footprints and zero memory leaks. The firmware was compiled using the official **PlatformIO ESP-IDF / Arduino framework** targeting the Espressif ESP32 DevKit v4:

Resource Metric	Measured Utilization	Hardware Budget / Limit	Utilization Percentage	Engineering Assessment
RAM (Internal SRAM)	49,424 Bytes	327,680 Bytes	**15.1 %**	**PASS** Massive headroom (84.9% free) for dynamic queues and buffers.
Flash Memory (ROM)	1,047,106 Bytes	1,310,720 Bytes	**79.9 %**	**PASS** Fits comfortably within standard app partition; leaves 263 KB free.
FreeRTOS Task Workers	12 application tasks	Core 1 Dedicated	**100 % Core 1**	**PASS** Core 0 dedicated to Wi-Fi/lwIP; 0 scheduling conflicts.
Hardware Watchdog (TWDT)	8000 ms panic threshold	6000 ms Heartbeat	**100 % Supervised**	**PASS** Prevents system lockup with automatic fail-safe reset.
Blocking Delays	0 ms (Zero `delay()`)	0 ms allowed	**0.0 %**	**PASS** 100% non-blocking; FreeRTOS tick yields throughout.

8. Known Architectural Limitations & Design Notes

In compliance with rigorous engineering documentation standards, the following six design trade-offs are explicitly noted:

1. **Multiplexed Event Feed (Adafruit IO Quotas):** Adafruit IO enforces a strict limit of 10 feeds on free accounts. The firmware utilizes 6 feeds for telemetry (Heart Rate, SpO2, Body Temp, Room Temp, Room Oxygen, AQI), 3 feeds for bi-directional control (Dosage, Bed Angle, Sampling Rate), and multiplexes all clinical alarms, facility warnings, and 60-second summary reports onto the single shared `hospital-events` topic.
2. **Logical Role Separation:** The segregation between Medical Staff and Facility Management dashboards is implemented at the topic and dashboard layout level rather than via cryptographic multi-tenant access control (RBAC).
3. **Replayed Telemetry Arrival Timestamps:** Offline records replayed to Adafruit IO carry the broker reconnection timestamp rather than original sampling time due to standard Adafruit IO MQTT ingestion constraints and the absence of a battery-backed physical RTC.
4. **Bounded FIFO RAM Buffer:** The offline buffer utilizes an in-memory FreeRTOS queue holding 64 records (~36.8 KB SRAM). When capacity is exceeded during extended outages, newest records are dropped to preserve early failure data. Data is volatile across power cycles.
5. **Task Watchdog & SYSTEM_DEGRADED Window:** `SYSTEM_DEGRADED` indicates a worker task starved its heartbeat while transport remained active. The hardware TWDT enforces an 8000 ms panic threshold; if a task hangs permanently, the chip resets ~2 seconds after entering `SYSTEM_DEGRADED`.

6. **Simulated Sensor Telemetry:** In Wokwi simulation mode, PPG, ECG, and ambient room parameters are generated via mathematical models and UART CLI overrides.

9. Verification, Test Suite & Simulation Guide

9.1 Automated PlatformIO Unity Unit Tests

The project incorporates automated unit testing located under `test/test_main.cpp`. Executed via `pio test -e esp32dev --without-uploading --without-testing`, the suite verifies **15 unit test cases**:

- Strict numeric string parsers verifying rejection of `NaN`, `Inf`, empty strings, trailing alphabetic garbage, and negative boundary inputs.
- Medication dosage safety clamping validating that requests > 80 mg/hr are restricted and require authorization.
- Slew-rate limiter interpolation bounds confirming gradual mechanical transitions.

9.2 Interactive Serial Simulation CLI Commands

An interactive Serial CLI (115200 baud) is embedded directly into the firmware to allow evaluators to inject physiological anomalies and simulate network disconnects in Wokwi:

Command Syntax	Target Subsystem	Simulated Physiological / System Effect
`SET_HR 130`	Task 1 & Task 2	Forces acute tachycardia (130 BPM); triggers CRITICAL red alarm and auto-elevates bed to 90°.
`SET_SPO2 90`	Task 1 & Task 4	Forces respiratory hypoxia (90% SpO2); triggers Task 4 auto-elevation to 45° Breathing Support.
`SET_TEMP 39.8`	Task 1 & Task 2	Forces high-grade fever (39.8 °C); triggers clinical warning alert and 5s adaptive sampling.
`RESET_VITALS`	Task 1	Clears all overrides and resumes nominal random physiological vitals.
`SET_ROOM 35`	Task 2 Facility	Forces ambient room temperature to 35.0 °C; emits facility compliance warning.
`SET_AQI 180`	Task 2 Facility	Forces hazardous air quality (180 AQI); triggers ventilation warning on `hospital-events`.
`SIM_OFFLINE WIFI`	Task 6 Offline	Disconnects Wi-Fi; OLED displays `LOGGING OFFLINE`; triggers rapid double-pip buzzer; buffers into FIFO.
`SIM_OFFLINE MQTT`	Task 6 Offline	Disconnects MQTT socket; triggers slow single-beep buzzer; buffers telemetry into RAM FIFO.
`SIM_ONLINE`	Task 6 Sync	Restores connectivity; mutes buzzer; executes exponential backoff replay of all 64 FIFO records.

9.3 How to Run in Browser Wokwi

1. Navigate to the live Wokwi project: **https://wokwi.com/projects/479638666615150593**.
2. Click the green **Play** button in the top toolbar to start the simulation.

3. Observe the **SSD1306 OLED** display rendering live medication dosage and the bottom offline ticker.
4. In the Serial Monitor, type `SET_SPO2 90` to watch the **SG90 Servo** articulate smoothly to 45° for breathing support.
5. Type `SIM_OFFLINE WIFI` to observe the persistent `[LOGGING OFFLINE]` banner and acoustic double-pip cadence.
6. Type `SIM_ONLINE` to observe automatic background reconnection and FIFO replay.

10. Conclusion & Evaluation Readiness

The Smart Hospital IoT System demonstrates production-oriented embedded firmware design. By decoupling network I/O from clinical worker tasks using FreeRTOS Symmetric Multiprocessing on the ESP32, the system guarantees that high-priority medical safety loops remain responsive under all operating conditions. With non-blocking peripheral drivers, dual-tier watchdog supervision, strict two-factor medication interlocks, intelligent autonomous bed articulation, and resilient offline replay buffers, the project fully satisfies all engineering criteria established by the Elevance Skills IoT Internship Program.

> **Final Submission Verification Summary**
>
> • **GitHub Repository:** https://github.com/iamsayanghosh/Smart-Hospital-IoT (100% clean history, 0 CI commits)
> • **Live Wokwi Simulation:** https://wokwi.com/projects/479638666615150593 (All 34 modules loaded and wired)
> • **Build Metrics:** 0 errors, 0 warnings, RAM 15.1%, Flash 79.9%, zero blocking delay() calls.